

BUSINESS REQUIREMENTS DOCUMENT (BRD)

PROJECT DETAILS

PROJECT NAME		
Netflix Content Recommender System		
CREATOR		
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1. EXECUTIVE SUMMARY SNAPSHOT:

Netflix aims to enhance user experience by delivering personalized content recommendations based on users' viewing preferences. The purpose of developing this project is to utilize a hybrid-based content recommendation technique in order to provide accurate, relevant, and personalized content suggestions.

This solution is intended to improve content discovery, increase user retention, encourage longer viewing sessions, and support Netflix's business objective of delivering an engaging experience.

This document defines the business objectives, project scope, stakeholder requirements, business processes, constraints, assumptions, and success criteria required for implementing the development and integration of the recommendation system to the platform. It serves as the foundation for aligning business stakeholders and the development team throughout the project lifecycle.

2. PROJECT DESCRIPTION:

Netflix is one of the world's leading OTT platforms serving millions of users across multiple countries with an extensive library of movies, TV shows, documentaries, and original content. As the volume of content increases, there is a crucial need to develop and integrate a recommendation system into the platform that generates content recommendations for all users based on their preferences.

The type of recommendation model will be hybrid-based (content-based + collaborative-based). By providing accurate and timely recommendations, the system is expected to increase user engagement, improve customer satisfaction, reduce content discovery time, and enhance subscriber retention.

The proposed solution will integrate seamlessly with the existing Netflix platform through secure APIs and cloud-based infrastructure. It will support scalability, high availability, and continuous model improvements while maintaining compliance with organizational security and data privacy standards.

3.

PROJECT SCOPE	
The project scope defines the boundaries of the Netflix Hybrid Content Recommendation System by identifying the functionalities, deliverables, and activities that are included within the project, as well as those that are excluded. Clearly defining the project scope helps establish stakeholder expectations, prevents scope creep, and ensures that project resources remain focused on achieving the intended business objectives.	
IN-SCOPE ITEMS	OUT-OF-SCOPE ITEMS
Collection and processing of user interaction data, including viewing history, ratings, and watchlist activity.	Customer support operations.
Processing and management of movie and TV show metadata.	Marketing and promotional campaigns.
Ranking and prioritization of recommended content.	Video streaming and playback services.
Integration of the recommendation engine with the Netflix application through secure REST APIs.	Recommendation of advertisements or promotional content.
Deployment of the recommendation model within Netflix's approved cloud infrastructure.	Subscription and payment management.

4.

BUSINESS DRIVERS	
The business drivers describe the key organizational goals and strategic objectives that justify the implementation of the Netflix Hybrid Content Recommendation System. These drivers highlight the business value expected from the solution and provide the foundation for defining project priorities, requirements, and success criteria.	
Business Driver 1:	Improve personalized content discovery by delivering highly relevant recommendations tailored to each user's preferences.
Business Driver 2:	Increase customer engagement by encouraging longer viewing sessions

	through accurate recommendations.
Business Driver 3:	Improve subscriber retention by reducing the effort required to discover relevant content.
Business Driver 4:	Leverage user behavior and content metadata to continuously improve recommendation quality and support data-driven business decisions.

5. PRESENT PROCESS:

Currently, the brand utilizes a content-based recommendation system. This has one major issue: the model doesn't work for any new title that it hasn't seen in its training phase. As a result, the system fails with any new title that didn't exist in the dataset (on which the model was trained) previously.

Current Business Process:

1. User logs into the application.
2. Selects their profile
3. Browse for suggested content on the homepage (based on watch history) or manually by entering the content title as a query in the search box.
4. Gets the results and similar titles to their selected title based on existing content data.
5. The user selects and watches the recommended content.
6. All user interactions with the recommended content are recorded in a log file for future recommendations.

Current Business Challenges:

1. Limited ability to handle newly added content (cold-start problem).
2. Recommendations primarily rely on content similarity and may not adequately reflect changing user preferences.
3. Recommendation accuracy decreases when user interaction history is limited.
4. Existing recommendations do not improve personalization.
5. Reduced recommendation quality may impact user engagement and content discovery.

6. PROPOSED PROCESS:

Hence, there is a new requirement to develop a hybrid-based content recommendation model that improves recommendation quality while supporting scalability, high availability, and seamless integration with the existing platform.

Proposed Business Process:

1. The user logs into the Netflix application.
2. The user selects a profile.
3. The recommendation service retrieves the user's viewing history, ratings, watchlist information, and profile preferences.

4. The system retrieves relevant movies and TV shows from the content repository.
5. The newly designed recommendation engine analyzes both user behavior and content characteristics to generate personalized recommendations.
6. The system ranks the recommended content based on predicted user relevance.
7. Personalized recommendations are displayed on the Netflix home page and related content sections.
8. The user selects and watches the recommended content.
9. User interactions are continuously recorded and used to improve future recommendations.
10. The recommendation model is periodically updated using newly available user interaction data and content metadata to continuously improve recommendation quality.

Expected Business Benefits:

1. Improved recommendation accuracy and personalization.
2. Faster and more efficient content discovery.
3. User churn rate decreases.
4. Better handling of newly added content and cold-start scenarios.
5. Increased diversity and relevance of recommended content.

7. FUNCTIONAL REQUIREMENTS:

The following requirements specify the core capabilities and business functions that the Hybrid Content Recommendation System shall implement to deliver accurate and personalized content recommendations.

PRIORITY		
The following section presents the requirements prioritized according to their business importance and implementation priority. This prioritization assists stakeholders in planning, development, testing, and project execution.		
VALUE	STATUS	DESCRIPTION
1	High	The system shall process and utilize content metadata (e.g., genre, cast, director, keywords, description) to generate recommendations.
2	Medium	The system shall rank recommended content based on relevance and display the Top-N recommendations.
3	High	Recommendations should be generated based on similar users. Hence,

		following collaborative recommendations.
4	High	The system shall notify the requesting application when recommendations cannot be generated due to missing or invalid input data.
5	High	The system shall maintain logs of generated recommendations for performance analysis and auditing.
6	Low	The system shall automatically update recommendations when user preferences or interaction history change.
7	Low	The system shall log recommendation requests, system errors, and operational events for monitoring and troubleshooting purposes.

CATEGORIES (RC1)			
The following table categorizes the requirements according to their business purpose, implementation priority, and the stakeholder who raised the requirement. This classification improves requirement traceability, prioritization, and stakeholder communication throughout the project lifecycle.			
ID	REQUIREMENT	PRIORITY	RAISED BY
RC-01	Content-Based Recommendation using movie metadata	High	Product Manager
RC-02	Recommendation Ranking (Top-N Recommendations)	Medium	Product Manager
RC-03	Collaborative	High	Data Science / ML

	Recommendation using similar users		Engineers
RC-04	Error Handling for Recommendation Requests	High	Software Engineering Team
RC-05	Recommendation Logging for Auditing and Performance Analysis	High	Software Engineering Team
RC-06	Recommendation results refresh automatically based on user activity	Low	Product Manager
RC-07	System Logging and Monitoring	Low	DevOps Team

8.

NON-FUNCTIONAL REQUIREMENTS	
The following requirements define the quality standards and operational characteristics required to ensure the recommendation system is secure, reliable, scalable, efficient, and maintainable.	
ID	REQUIREMENT
1.	The recommendation system shall provide high-performance, scalable, highly available, and fault-tolerant recommendation services while efficiently utilizing computing resources to ensure consistent response times under varying workloads.
2.	The recommendation system shall ensure safe, reliable, and uninterrupted operation by preventing invalid recommendations, maintaining data integrity, and minimizing the impact of system failures.
3.	The recommendation system shall protect user data and system resources through authentication, authorization, encryption, secure communication, access control,

	and compliance with organizational security and privacy policies.
4.	The recommendation system shall deliver reliable, scalable, maintainable, efficient, and user-friendly recommendation services while ensuring high availability, portability, and testability throughout its lifecycle.

9.

GLOSSARY	
These are the following terminologies and abbreviations I used throughout this document:	
TERM/ABBREVIATION	EXPLANATION
Top-N Recommendations	A ranked list of the most relevant N recommendations presented to the user.
Content-based Recommendation System	A recommendation technique that suggests content similar to what a user has previously watched based on content attributes.
Collaborative Recommendation System	A recommendation technique that recommends content based on the preferences and behaviors of similar users.
REST API	An API that follows REST architectural principles for communication between applications.
Hybrid Recommendation System	A recommendation approach that combines Content-Based Filtering and Collaborative Filtering to improve recommendation accuracy.
Content Metadata	Information describing datasets' structure.

10.

REFERENCES
These are the following resources I used to write this BRD documentation:

NAME	LOCATION
BRD Template by Smartsheet	https://www.smartsheet.com/content/business-requirement-document-templates
ChatGPT	https://chatgpt.com/

11. APPENDIX:

STAKEHOLDER LIST	
Stakeholder	Role
Product Manager	Defines business objectives and product vision.
Business Analyst	Gathers, analyzes, and documents business requirements.
Data Scientist	Develops and validates the recommendation model.
ML Engineer	Assists data scientists in building the model (if necessary in any case) and deploys and maintains it.
Data Engineer	Builds and maintains data pipelines.
Software Engineer	Integrates the recommendation service with the Netflix platform.
QA Engineer	Tests the recommendation system against business requirements.
DevOps Engineer	Deploys, monitors, and maintains the production environment.

SUCCESS METRICS	
KPI	Target
Recommendation Response Time	< 2 seconds
System Availability	99.9% or above
API Success Rate	99% or above
Recommendation Accuracy	Improved over current system

User Engagement	Increase in viewing sessions
Subscriber Retention	Increase after deployment